to certain diseases simply because their normal body temperature inhibits growth of the pathogens. Gastric acidity is an innate physiologic barrier to infection because very few ingested microorganisms can survive the low pH of the stomach contents.

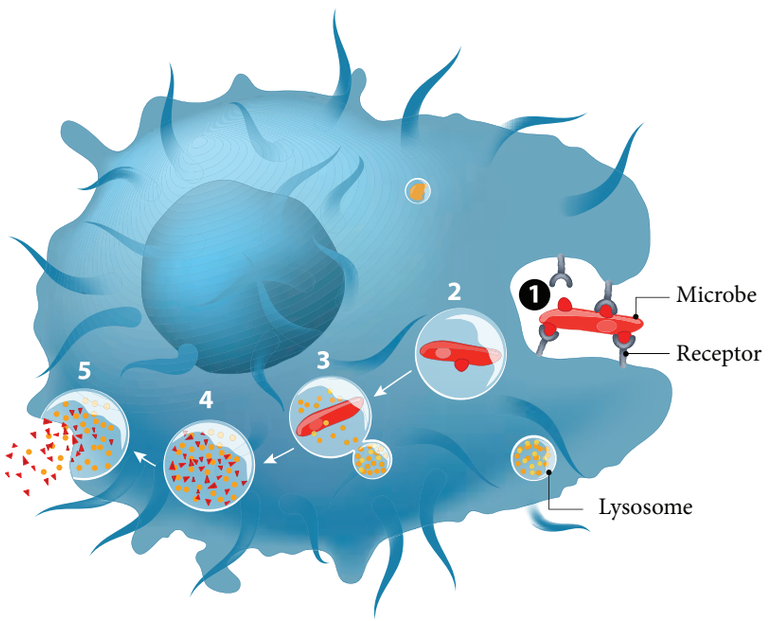
A variety of soluble factors contribute to innate immunity, among them are the soluble proteins lysozyme, interferon, and complement. Lysozyme, a hydrolytic enzyme found in mucous secretions and in tears, is able to cleave the peptidoglycan layer of the bacterial cell wall. Interferon comprises a group of proteins produced by virus-infected cells which prevents viral infection of neighbouring cells. Complements lyse bacteria by forming membrane attack complex.



The newborn babies (neonates) are susceptible to some infections that do not affect adults. The reason is that their stomach contents are less acidic i.e higher pH than those of adults. The pH variation provides optimum medium for the growth of pathogenic microorganisms.

Phagocytic Barriers

Another important innate defense mechanism is the ingestion of extracellular particulate material by phagocytosis. The process of phagocytosis was discovered by Metchnikoff. The term phagocytic denotes the engulfment and digestion of whole cells. The two major cell types in the body which are associated with the engulfment and digestion of microorganism are the polymorphonuclear leucocytes and the macrophages. Minor cell types are the eosinophils. The process of phagocytosis involves the following steps:



1. Binding and absorption
2. Phagosome formation
3. Phagosome and lysosome fuse to form a phagolysosome
4. Digestion
5. Release of microbial products

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